

# Vibing Art: AI-Generated Interactive Visuals Reacting to Sound and Movement in Your Pocket

Mingxuan Xie  
Vrch Limited  
London, United Kingdom  
ming@vrch.io

Bingyu Yu  
Vrch Limited  
Shanghai, China  
bingshu@vrch.io

Tianzi Hou  
Vrch Limited  
Edinburgh, United Kingdom  
tianzi@vrch.io

Yunyi Chen  
Vrch Limited  
Shanghai, China  
yunyi@vrch.io

## Abstract

Creating real-time, audio-reactive visuals has long demanded hand-coded shaders and heavyweight desktop VJ software, which keeps live audiovisual performance out of reach for most musicians, dancers, and casual creators. Vibing Art collapses this entire pipeline onto a single iPhone. A user describes the visual they want in plain language and in about fifteen seconds, the app returns a live visual, rendered on the device, that reacts to sound, hand gestures, body pose, and the camera content. The user can refine each piece simply by continuing the conversation, then mirrors it to any external display and shares it as a .vibe file that friends can remix and jam on. Vibing Art is a simplified, self-contained mobile solution of our Agentic VJ System, carrying interactive audiovisual creation from the professional stage into everyone's pocket.

## CCS Concepts

• Human-centered computing → Interactive systems and tools; Ubiquitous and mobile computing systems and tools; • Computing methodologies → Artificial intelligence; Natural language generation.

## Keywords

generative AI, real-time graphics, shader synthesis, audio-reactive visuals, on-device machine learning, mobile, VJ, live performance, creativity support tools

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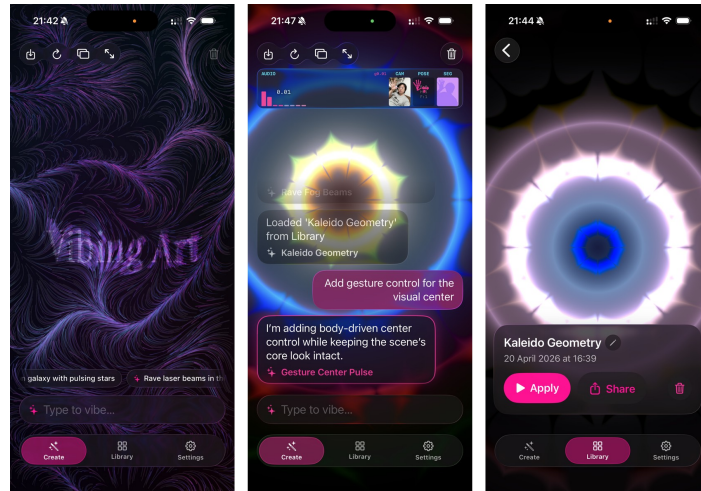


**Figure 1: Vibing Art jamming with friends. Each player's iPhone or iPad shows a different visual generated from each player's prompt while all reacting to the same music playing in the room.**

## 1 Introduction

Real-time, audio-reactive visuals — the glowing geometry behind a DJ set, the pulsing backdrop of a live stream, the reactive projections at a club night — have become a familiar part of contemporary performance scene. Yet creating them is still largely the preserve of specialists. Traditional VJ practice combines pre-rendered CGI assets, VFX algorithms with heavyweight desktop software, MIDI control surfaces, and workstation grade GPUs. A newcomer who wants original, reactive visuals usually has to learn professional desktop software, shader programming, audio beat detection logic, and projection setup all at once, spend weeks assembling a stage-ready set, and then carry dedicated hardware to every venue. As a result, only a small fraction of the musicians, performers, and streamers who would benefit from interactive visuals ever get to create their own.

Vibing Art removes these barriers by collapsing the whole pipeline onto a device that is already in the user's pocket. The user simply describes the visual they have in mind in everyday language, and within roughly fifteen seconds the app renders a live, audio-reactive visual directly on the phone. Iteration is also conversational: a follow-up message such as “make it black and white”, “slow the rotation down”, or “let my right hand control the centre” refines the visual that is already on screen instead of regenerating it from scratch. The finished result can be driven in real time by sound, gesture, body, and camera stream, mirrored to any external display,



**Figure 2: The Vibing Art user journey on iPhone. Left: prompt entry with one-tap inspiration cards. Centre: a generated visual driven live by the natural-language edit “add gesture control for the visual center”. Right: a saved piece in the personal library, ready to apply, share, or remix as a .vibe file.**

and shared as a remixable file that other people can open and build on.

Vibing Art is a deliberately simplified, fully self-contained mobile distillation of our *Agentic VJ System* [Xie and Hou 2025], a multi-agent, workstation-class generative visual performance system that we demonstrated at SIGGRAPH Asia 2025 Real-Time Live!. Where that system targets professionals with a fully local AI inference pipeline and professional audio and video I/O, Vibing Art reduces the same core idea — turning natural language into live, reactive graphics — to a single conversation on a consumer device. It deliberately trades breadth of professional control for immediacy and accessibility, so that anyone, with easy home setup, can make and perform interactive visuals in seconds (Fig. 1).

## 2 Interacting with Vibing Art

Vibing Art presents creation as a conversation. The app turns each prompt into a live visual that the user can then play with, project onto a larger screen, and pass on to others. Three capabilities define the experience (Fig. 2).

### 2.1 Conversational creation and editing

Every piece begins from a single prompt, and a row of one-tap inspiration cards lets a first-time user start with no typing at all (Fig. 2, left). Each visual is backed by a freshly generated program rather than a fixed preset, follow-up prompts edit the existing piece — adjusting its palette, motion, composition, or which sensor drives which parameter — instead of starting over. This keeps the user “in the vibe”, rather than breaking concentration to hunt through menus and sliders.

### 2.2 Real-time multimodal reactivity

Once a visual exists, it can respond, frame by frame, to four streams of sensor data that are all analysed on the device (Fig. 2, centre). The audio stream provides an eight-band frequency spectrum together

with beat detection and a coarse classification of voice, percussion, and timbre. The camera contributes full-body pose with thirty-three skeletal keypoints, independent tracking of the left and right hands with pinch, open, and point gestures, and real-time human segmentation that can composite the performer directly into the scene. A small live overlay shows which of the audio, camera, pose, and segmentation channels are currently active, and a natural-language edit can rebind any of them — for instance, “let my hand control the centre” — turning what was a passive animation into an instrument the user plays with voice and body.

### 2.3 Save, share, and remix

A finished visual is a pure, full-screen render, which turns the iPhone into a self-contained VJ machine. Its output can be sent over AirPlay or a USB-C / HDMI cable to a television, projector, or LED wall, and the rendering canvas is rebuilt automatically for screen ratios detected from targeted displays so that nothing is stretched or awkwardly cropped (Fig. 3). Each piece can be saved to a personal library and can be exported as a .vibe file that packages the generated visual together with its originating prompt and its reactivity bindings. A recipient can apply that .vibe and immediately keep editing it, so a visual made by one person becomes a starting point that others remix.

## 3 System Architecture

Vibing Art runs on a hybrid architecture that divides the work according to its latency budget: the heavy generative reasoning happens in the cloud, while every signal that must keep pace with the frame rate is processed on the phone.

When a prompt arrives, our cloud-side code-generation agent turns it into a custom shader program. The agent pairs a fine-tuned shader-generation model with a runtime harness that validates the generated code, executes it in a sandbox, and applies error-category-directed automatic repair whenever something fails. Responses are



**Figure 3: Playing with an external display. The generated visual is mirrored from the device to a large screen over AirPlay, and the player steers it live with a raised hand through the camera-based pose and gesture tracking.**

streamed and use prefix caching, so the first bytes come back in seconds, and the generation loop as a whole reaches a success rate above 95% at an average of roughly fifteen seconds per visual — fast enough to keep a user engaged and experimenting.

On the device, all of the real-time sensing is handled by Apple’s Core ML and Vision frameworks [Apple Inc. 2024a,b], which perform the audio analysis, the body-pose and hand-gesture tracking, and the human segmentation from the camera feed. The generated

visuals themselves run at 60 frames per second, with end-to-end sensor-to-render latency under 30 milliseconds. Because no per-frame signal ever leaves the phone, reactivity stays low-latency and private, and a visual keeps working offline once it has been generated; only the one-off generation step needs a network connection.

## 4 The Appy Hour Experience

At our Appy Hour station, attendees create a working visual from scratch in well under a minute. A visitor types a prompt, watches the visual stream into existence and then drives it live — clapping or playing music around to trigger the beat detector, raising a hand to steer the composition, or active the front camera to be composited into the scene. With a single tap the visual is mirrored to the booth’s large display (Fig. 3), and visitors can AirDrop a .vibe of their creation to a friend, or to their own phone, and remix it on the spot. The message we hope attendees take away is simple: with a single sentence, anyone can become an interactive-media artist and carry the whole creative studio in their pocket.

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